

32 The Segregation and Maximum Mixedness Models

When we classify a reactor as ideal what we are saying is that the contents are *well mixed*

The degree of mixing is important, because it defines the *concentration* which defines the *rate*

The rate then determines the *conversion*

The exception is for *first order reactions*, where the conversion does not depend on the concentration

Mole balances (in terms of conversion) for CSTR, PFR, and batch reactor for a first order reaction. Assuming constant volumetric flowrate:

- CSTR: $V = \frac{v_0 X}{k(1-X)}$
- PFR: $\frac{dX}{dV} = \frac{k(1-X)}{v_0}$
- Batch: $\frac{dX}{dt} = k(1-X)$

Real reactors in general are not well mixed. Hence for non-first order reactions, we need methods for determining the concentration *over the entire volume of fluid*

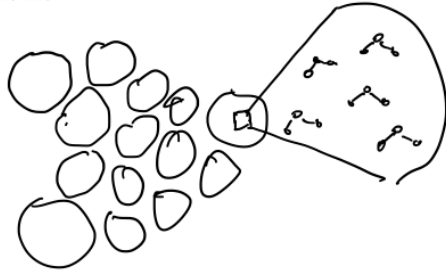
Ideally, there would be a function to tell us this, but that is not reasonable. Instead, we use *models*

These models differ in how they treat mixing amongst molecules of different "ages"

In mixing models, "age" means *the amount of time that a molecule has been inside the reactor*

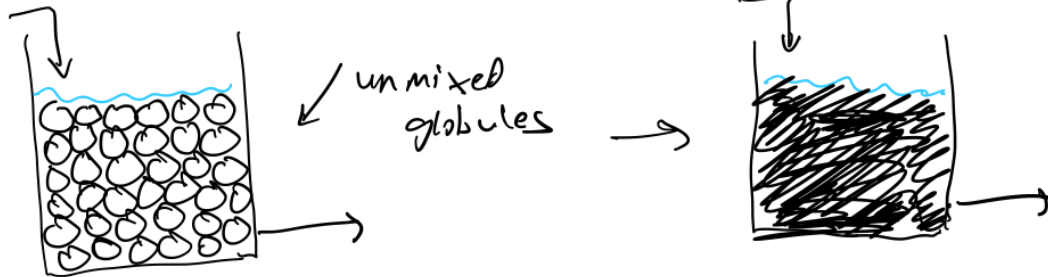
"Groups" of molecules of the same age are called *globules*

We can draw a globule like



Each globule contains molecules of the same age
But all globules are different ages

Globules within a reactor might look like (remember this is a model...)



There are a variety of ways to model mixing in reactors. In this class, we will focus on the

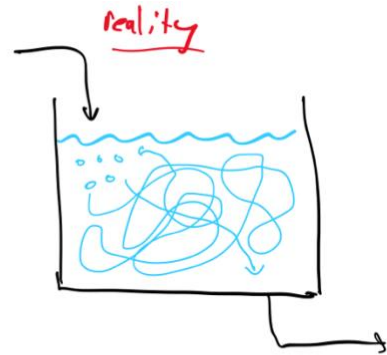
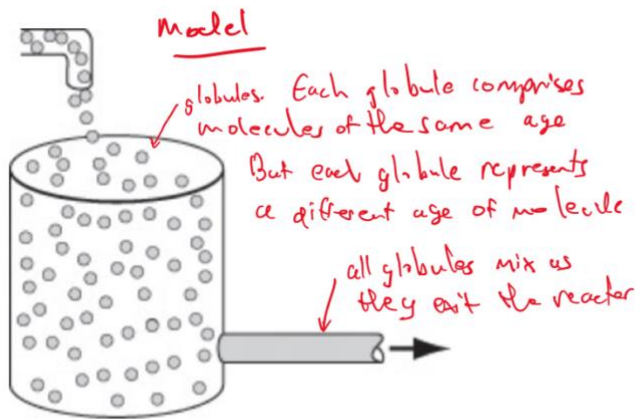
segregation model
maximum mixedness model

- Segregation model: Globules do not mix with other globules until they exit the reactor
 - Maximum mixedness model: Globules mix with other globules as soon as they enter the reactor
- In this model, the fluid is completely mixed at the molecular level

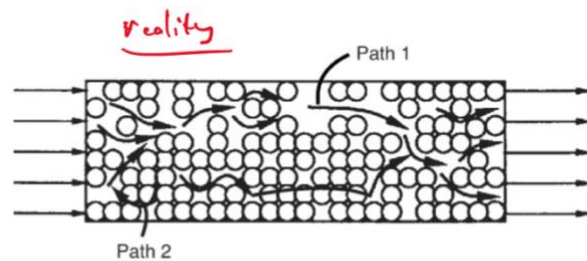
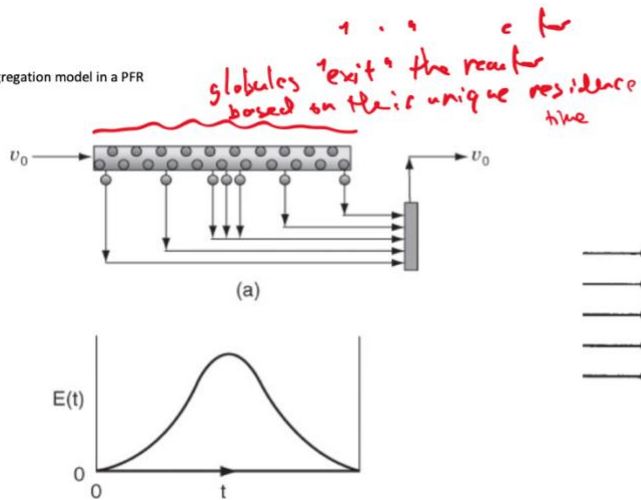
For single reactions with orders greater than one or less than zero, the segregation model will predict the highest conversion. For reaction orders between zero and one, the maximum mixedness model will predict the highest conversion.

For reactions with orders of one both models predict the same conversion

- Segregation model in a CSTR



- Segregation model in a PFR



In the segregation model, there is no mixing / flow between globules, and hence each globule can be modeled as a batch reactor

The mean conversion is given by:

Mean conversion of globules spending between time t and $t + \Delta t$ in the reactor = Conversion achieved in a globule spending time t in the reactor $\times$ Fraction of globules that spend between time t and $t + \Delta t$ in the reactor. Mathematically, this looks like

$$d\bar{x} = x(t) E(t) dt$$

Rearranging: $\frac{d\bar{x}}{dt} = x(t)E(t)$. Integrating:

$$\bar{x} = \int_0^{\infty} x(t) E(t) dt$$

determined from the batch reactor mole balance

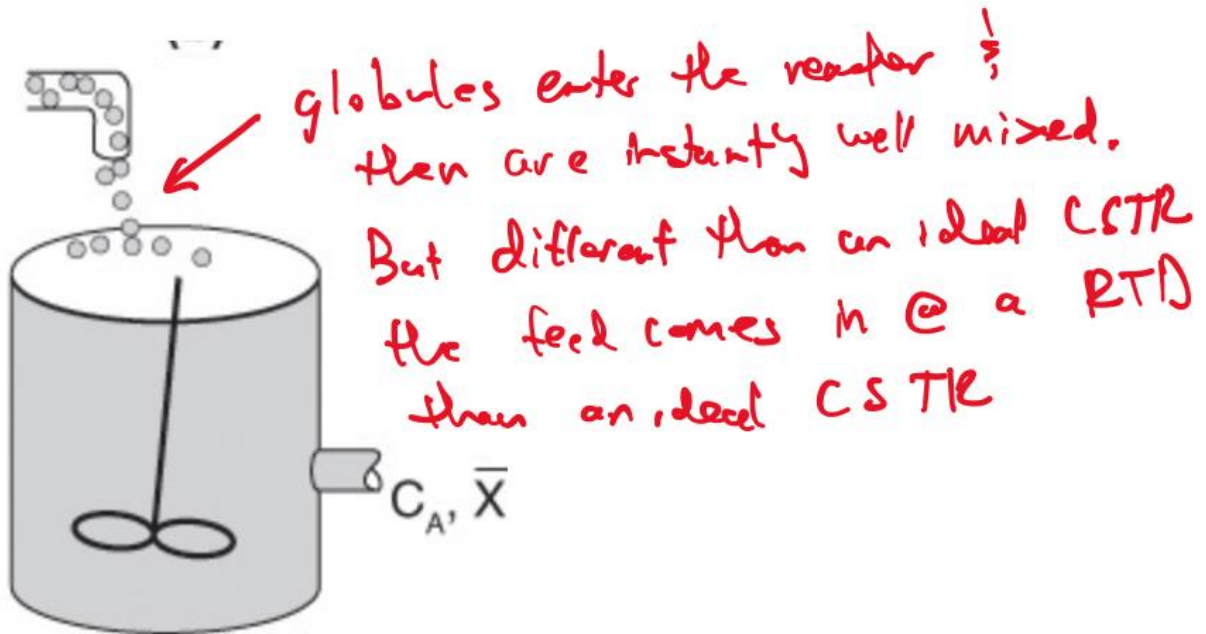
$$\frac{dx}{dt} = \frac{-r_A V}{N_{A0}}$$

Thus, if we have the rate law and the RTD ($E(t)$), we can find the mean conversion.

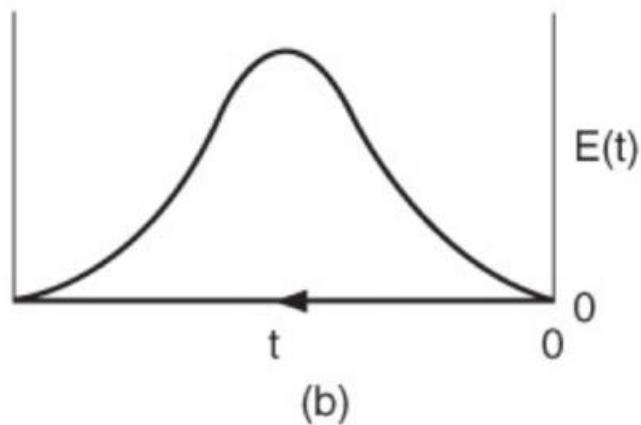
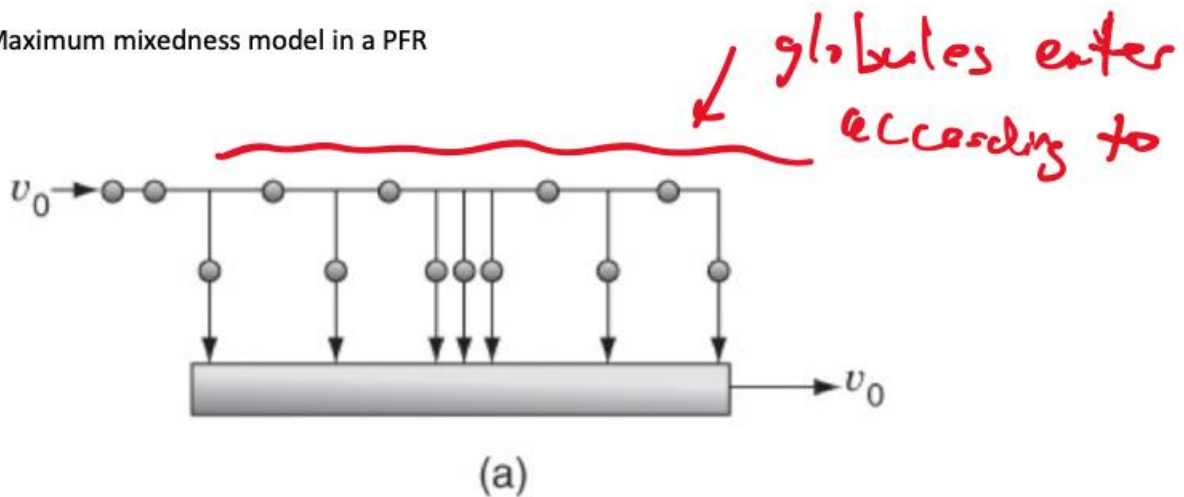
In the maximum mixedness model, molecules mix as soon as they enter the reactor

- Maximum mixedness model in a CSTR

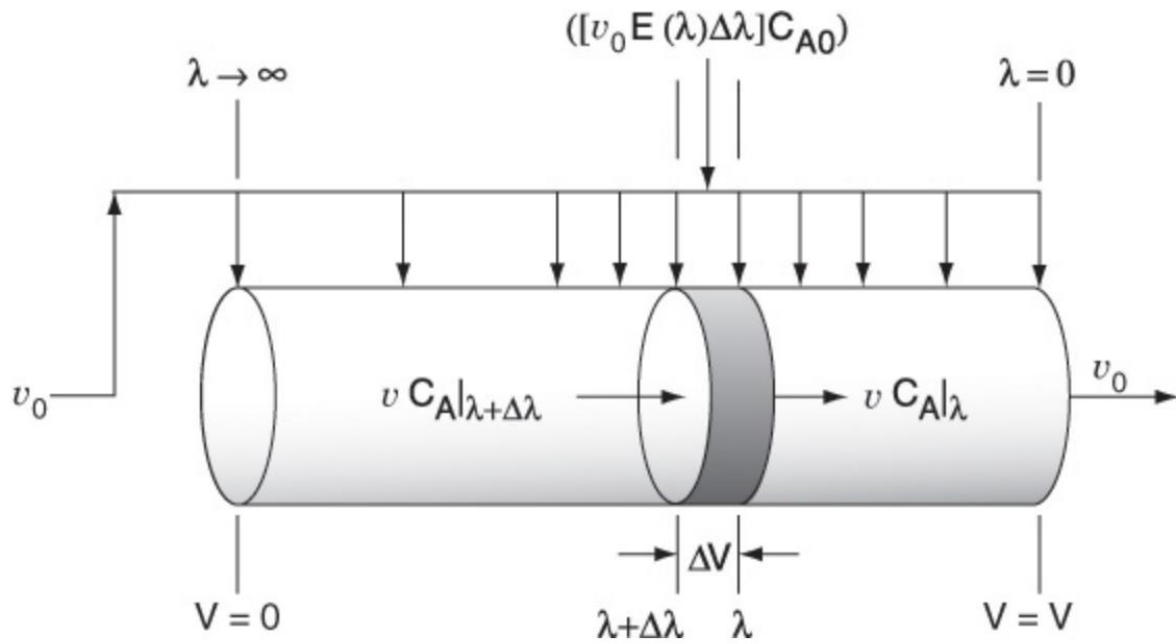
- Maximum mixedness model in a CSTR



- Maximum mixedness model in a PFR



The derivation for the mean conversion under the maximum mixedness model involves a lot of math that I am going to skip. But you can find it [here](#). Basically, we define a new variable λ , which is the time it takes a molecule to move to the end of the reactor.



We then do a mole balance on species A in this reactor model, and arrive at $\frac{dX}{dz} = -\frac{r_A}{C_{A0}} - \frac{E(\bar{T}-z)}{1-F(\bar{T}-z)} X$, where z is the difference between the longest time measured in the $E(t)$ curve, $\bar{T}$ and λ . I.e., $\lambda = \bar{T} - z$.